NFL Game Simulator v4: Settings, Data Points, and Referee Names

Sep 28, 2026 · @Brayden

Quick start

Edit these lines in step 2 of the notebook, then run the cell. Everything left as `None` falls back to the data: depth-chart QB, injury reports, stadium weather average, and the referee listed in the schedule once assigned.

```
HOME, AWAY, NEUTRAL = "CHI", "NYJ", False
HOME_QB, AWAY_QB = None, None          # e.g. "Case Keenum" or "C.Keenum"
OUT = []                               # e.g. ["T.Bagent", "Joe Thuney"]
WEATHER = None                         # e.g. {"temp": 38, "wind": 18}
REFEREE = None                         # e.g. "Shawn Hochuli" or "Hochuli"
```

Setting	What to type	Example	Notes
`HOME`, `AWAY`	Team abbreviation, in quotes	`"CHI"`	See Team abbreviations below
`NEUTRAL`	`True` or `False`	`True`	`True` for London, Germany, Brazil, Super Bowl. Removes home field
`HOME_QB`, `AWAY_QB`	`None` or a QB name	`"Case Keenum"`	Full name or first-initial format both work
`OUT`	List of player names	`["T.Bagent", "Joe Thuney"]`	Rules players out on top of the injury report. Works for QBs, WR/TE/RB, OL and defense
`WEATHER`	`None` or temp (F) and wind (mph)	`{"temp": 38, "wind": 18}`	Ignored for domes and closed roofs
`REFEREE`	`None` or crew chief name	`"Hochuli"`	Full name or last name, any capitalization

Setting	What to type	Example	Notes
`SEASON`, `WEEK`	Numbers	`2026`, `4`	Which week's rosters and injury report to use
`AS_OF`	Date in quotes	`"2026-09-29"`	Ratings use games before this date. Set it to the day after the last game you want counted
`n_sims`	Number	`20000`	More = smoother percentages, slower. 10,000-20,000 is plenty
`features`	Tuple of factor names	`ALL_FEATURES`	Leave out a name to turn that factor off. See Model factors below

To turn one factor off for a single run:

```
no_h2h = tuple(f for f in ALL_FEATURES if f != "h2h")
simulate_game(model, HOME, AWAY, ctx=ctx, features=no_h2h)
```

Referee names

These 17 crew chiefs have worked games in 2025 or 2026 and are the valid `REFEREE` values. Type the full name or just the last name. Crew assignments are announced midweek, so leave `REFEREE = None` until then.

Crew chief	Games in data (2019-2026)	Flags per game	Away minus home flags
Alex Moore	19	13.8	+0.62
Adrian Hill	117	12.7	+0.62
Alex Kemp	120	12.6	+0.59
Shawn Hochuli	118	12.5	+0.46
Carl Cheffers	120	12.5	+0.48
Brad Rogers	111	12.4	+0.09

Crew chief	Games in data (2019-2026)	Flags per game	Away minus home flags
Clete Blakeman	120	12.3	+0.39
Ron Torbert	122	12.0	0.00
Land Clark	99	12.0	-0.03
Craig Wrolstad	120	12.0	+0.39
Clay Martin	116	11.8	+0.41
Scott Novak	115	11.8	+0.15
Shawn Smith	118	11.8	+0.59
Brad Allen	117	11.6	+0.62
Alan Eck	53	11.2	+0.23
John Hussey	122	11.2	+0.56
Bill Vinovich	121	11.1	+0.28

Flags per game are shrunk toward the league average, so crews with few games (Alex Moore) are pulled toward the middle. "Away minus home flags" above zero means the away team drew more flags in that crew's games; nearly every crew leans that way, which is the usual home-field effect.

Aliases handled automatically: "Bradley Rogers" = Brad Rogers, "Ronald Torbert" = Ron Torbert, "Adrian Hall" (a data typo) = Adrian Hill.

What the referee factor does today: nothing to the line. Across 2019-2026, a crew's flag rate did not predict total points and its home/away flag split did not predict the margin (both effects indistinguishable from zero), so the model shrinks both to 0.00. Setting `REFEREE` still prints the crew's tendencies, and the effect updates automatically if future data shows one. There is no public data grading whether calls were correct, so call accuracy itself can't be modeled.

Team abbreviations

Use these exact codes for `HOME` and `AWAY`. The Rams are `LA`, not `LAR`.

Code	Team	Code	Team
ARI	Arizona Cardinals	LA	Los Angeles Rams
ATL	Atlanta Falcons	LAC	Los Angeles Chargers
BAL	Baltimore Ravens	LV	Las Vegas Raiders
BUF	Buffalo Bills	MIA	Miami Dolphins
CAR	Carolina Panthers	MIN	Minnesota Vikings
CHI	Chicago Bears	NE	New England Patriots
CIN	Cincinnati Bengals	NO	New Orleans Saints
CLE	Cleveland Browns	NYG	New York Giants
DAL	Dallas Cowboys	NYJ	New York Jets
DEN	Denver Broncos	PHI	Philadelphia Eagles
DET	Detroit Lions	PIT	Pittsburgh Steelers
GB	Green Bay Packers	SEA	Seattle Seahawks
HOU	Houston Texans	SF	San Francisco 49ers
IND	Indianapolis Colts	TB	Tampa Bay Buccaneers
JAX	Jacksonville Jaguars	TEN	Tennessee Titans
KC	Kansas City Chiefs	WAS	Washington Commanders

Player name formats

Both formats work everywhere: the full name (`"Tyson Bagent"`) or the first-initial format the play-by-play data uses (`"T.Bagent"`). Capitalization doesn't matter; spelling does. Check spellings in `qb_table(model)` for QBs and `line_status(model, prep, "CHI", ctx)` for linemen and defenders.

Setting	Who it applies to	Example	What happens
`HOME_QB` / `AWAY_QB`	Any QB	`"Case Keenum"`	Forces him to start. No recent dropbacks = rated on his 2019-2024 history, or as a generic backup if none
`OUT`	QBs	`["T.Bagent"]`	The depth chart skips him and starts the next QB
`OUT`	WR, TE, RB	`["Puka Nacua"]`	His value above a replacement comes off the offense
`OUT`	OL and defensive starters	`["Joe Thuney", "M.Sweat"]`	Counts as a missing starter for his position group

Not overridable: kickers. The model uses the depth-chart kicker and moves to the next one only if the injury report or a reserve list has him out.

Suffixes: keep them exactly as the roster lists them (`"Michael Penix Jr."` , `"Anthony Richardson Sr."`).

Model factors

The model has 16 factors, all on by default. The "Key" column is the name to use in `features` . Every effect is estimated from data and shrunk by how reliable it is, so a factor with no measurable effect still shows up in the printout but barely moves the line.

Key	What it does	Data it uses	What it moves
`epa`	Team efficiency, opponent-adjusted, pass and run split	Play-by-play 2023-2026	The biggest driver of the line
`qb`	Starting QB vs the team's usual QB	QB EPA per dropback (+ 2019-2024 history)	Large when a starter is out

Key	What it does	Data it uses	What it moves
`skill`	WR, TE, RB value above a replacement, removed if out, moved on trades	Targets, carries, EPA	Up to ~1-2 pts for a star
`lines`	OL and defensive starters out vs the team's usual level	Snap counts, injury report, reserve lists	Missing OL starter: -0.028 EPA/play for his offense. Missing CB: +0.042 for the opponent. DL +0.016, LB +0.012, S ~0
`gameplan`	Pass rate vs defenses built like this opponent	Team game logs	Small
`style`	Over/under-performance vs similar defenses; current head coach's games count fully, older regimes 40%	Team game logs	Up to ~1 pt
`turnovers`	Turnover odds from QB mistakes and defensive takeaways	Interceptions, lost fumbles	Changes turnover counts and short fields
`fourth`	Coach go-for-it rate over expected	Every 4th down 2023-2026	Under 0.5 pt
`halftime`	Coach 2nd-half vs 1st-half swing	Half-by-half EPA	Shifts halftime score, not the final
`h2h`	Meetings in the last 2 years	Game results	Capped at 1.5 pts
`weather`	Wind over 10 mph, cold below 40F, indoor	Game conditions 2014-2026	Wind: -0.25 total pts per mph. Cold: -0.15 per degree. Indoor: +1.5
`rest_travel`	Rest days, short weeks, West Coast teams in early Eastern kickoffs	Schedule	West Coast early kickoff: about -1.3 pts

Key	What it does	Data it uses	What it moves
`pass_rush`	Pressure allowed vs pressure generated, log5 matchup	Sacks, QB hits	About 0.1 pt; already captured by team ratings
`kicker`	Depth-chart kicker's accuracy vs league at each distance	Every FG attempt	Small
`redzone`	TD vs FG finishing inside the 20	Drive results	Small, but helped in both test seasons
`referee`	Crew chief flag rate and home/away split	Penalties 2019-2026	Zero today (no measurable effect)

How player adjustments are weighted: QB, skill-player, line and matchup adjustments ride inside the team-rating blend at 35% weight. I tested full weight. It made 2025 predictions worse and pushed totals 1.5-6 points high, so single-player effects are noisier than they look and the model stays cautious about them.

Backtest with all 16 factors vs Vegas closing lines:

Test	Winner %: model / Vegas	Spread miss (pts): model / Vegas	ATS record	Over/under record
2026 Weeks 1-2	65.6% / 71.9%	11.9 / 11.6	15-16	14-18
2025 Week 4+	60.5% / 63.2%	10.3 / 9.9	112-111	115-109
2024 Week 4+	72.8% / 74.1%	10.0 / 9.3	99-122	120-101

Break-even against the spread at standard -110 odds is 52.4%. The model hasn't cleared it on spreads. Totals have been above 50% but not by enough to rule out luck.

Output fields and helper functions

`result = simulate_game(...)` returns a dictionary. Read any field with `result["field_name"]`.

Field	What it is
`home_win_pct`, `away_win_pct`, `tie_pct`	Share of simulations each outcome happened (0 to 1)
`proj_home_pts`, `proj_away_pts`	Average points scored
`proj_spread`	Model line, Vegas style: negative = home team favored
`proj_total`	Average combined points
`half_home`, `half_away`	Average halftime score
`poss_home`, `poss_away`	Average possessions per team
`to_home`, `to_away`	Average turnovers per team
`top_scores`	Five most common final scores, with percentages
`home_scores`, `away_scores`	Every simulated final score, for your own charts
`inputs`	Everything that went into the game: projected points, QB used, kicker, pressure rate, red-zone rate, starters out, weather, rest, referee

Function	What it does
`print_game(result)`	Readable summary of one game
`factor_breakdown(model, HOME, AWAY, ctx=ctx)`	Adds factors one at a time to show what moved the line
`simulate_week(model, SEASON, WEEK, ctx=ctx)`	Every game that week next to the Vegas line and total
`line_status(model, prep, "CHI", ctx)`	A team's OL and defensive starters, snap share, quality, and who's out
`qb_table(model)`	Active QBs: EPA per dropback and turnover rate
`coach_table(model)`	4th-down aggressiveness and halftime swing per head coach
`active_referees(model)`	Valid `REFEREE` names with tendencies

Function	What it does
`referee_table(model)`	All crew chiefs in the data, with their current effect on lines
`backtest(...)`	Replays past weeks using only data available before each game
`grade_vs_vegas(bt)`	Winner %, spread and total misses, ATS and over/under records vs Vegas
`ats_detail(bt)`	Game-by-game ATS and over/under picks from a backtest

Data files and weekly refresh

All data lives in `nfl_data/` next to the notebook and comes from nflverse. Delete a file and the next run downloads a fresh copy; files for finished seasons never need refreshing.

File	What's in it	Refresh?
`games.csv`	Schedule, scores, Vegas lines and totals, rest days, roof, weather, referee	Weekly
`pbp_2026.parquet`	Every 2026 play	Weekly
`injuries_2026.parquet`	Official injury report statuses	Weekly, after Friday's report
`roster_weekly_2026.parquet`	Rosters, including IR, PUP, NFI, suspended	Weekly
`depth_charts_2026.parquet`	Depth charts (starting QB, kicker)	Weekly
`snap_counts_2026.parquet`	Snap shares for every player (who starts on the line and defense)	Weekly
`pbp_2019` to `pbp_2025`, other past seasons	History for ratings, QB history, referees	Never

When things become known each week:

Day (Sunday game)	What's new	What to do
Tuesday-Wednesday	Last week's plays, snaps, rosters	Delete the weekly files above, re-run
Midweek	Referee crew assignments	Set `REFEREE`
Friday	Final injury report (Out / Doubtful / Questionable)	Delete `injuries_2026.parquet`, re-run with `carry_forward=False`
1-2 days before	Reliable weather forecast	Set `WEATHER` for outdoor games
90 min before kickoff	Inactives	Add surprise scratches to `OUT`

`carry_forward=True` (the notebook default early in the week) uses last week's injury report at reduced weight because the new one isn't out yet. Switch it to `False` once Friday's report is published so statuses count in full.